

Handout 2: Expanded Literature Review

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1 Student Outcomes

By the end of this activity, students will be able to:

1. Critically read and analyze research papers by identifying hypotheses, experimental methods, measurements, analyses, and conclusions.
2. Organize and summarize technical information from scientific literature in a structured and concise manner for future research use.
3. Evaluate the strengths, limitations, safety considerations, and validity of experimental studies presented in research papers.

2 Equipment

1. Use Google Scholar to download papers related to your project topic.
<https://scholar.google.com>
2. Accessing research papers.
 - (a) Students working on campus and connected to UIC WiFi can usually download papers directly through the university library subscriptions.
 - (b) Students working off campus can access papers using one of the following methods:
 - i. Use the UIC VPN service to connect to the university network. Instructions are available here:
<https://it.uic.edu/services/faculty-staff/uic-network/uic-vpn/>
 - ii. Use the UIC Library EZProxy service. Instructions are available here:
<https://library.uic.edu/help-articles/proxy-urls-ezproxy-2/>

3 Extensive Literature Review

You will now read 3 papers more carefully and make structured notes. Keep the notes concise, organized, and useful for future research and project development. For each paper, identify and summarize the following components.

1. **Hypothesis**

What is the main hypothesis or research question of the paper? What are the authors trying to test or demonstrate?

2. **Experiment**

What experiment or study was performed? Under what conditions was the experiment conducted?

Examples:

- Walking on a treadmill
- Walking at slow, normal, and fast speeds
- Walking on different incline or grade levels

3. **Measurements**

What variables or quantities were measured?

Examples:

- Speed
- Heart rate
- Muscle activity

What devices or sensors were used to obtain these measurements?

Examples:

- Microcontroller
- Heart rate sensor
- IMU sensor
- Motion capture system

Also identify where the sensors or devices were mounted on the subject or experimental setup.

4. **Analysis**

How was the collected data analyzed?

Identify:

- Metrics used for analysis
- Number and type of subjects
- Statistical tools or methods used

Examples:

- Average beats per minute over a 2-minute interval
- 10 healthy subjects
- Student t-test

- ANOVA

5. Safety and Ethics

Was the study approved by an Institutional Review Board (IRB)?

Do you anticipate any safety concerns or possible failures in the study? Did the authors mention any risks, limitations, or safety-related issues?

6. Results

What are the main findings of the paper? Which experiments, measurements, or analyses support these conclusions?

7. Limitations

What limitations are mentioned by the authors?

If limitations are not explicitly discussed, identify any potential weaknesses or limitations that you observe in the study.

Show your literature review to the teaching assistant.